ANNOTATING THE FOOD DOMAIN ACROSS DIFFERENT CONTEXTS

[Oct 16, 2026]

In this work, we propose annotation guidelines to capture food and its uses with a historical perspective. These guidelines are inspired by Frame Semantics, therefore, we briefly introduce here the concept that we borrowed from this theory for the present annotation guidelines.

Lexical Units and Frame Elements

Frame Semantics is a linguistic theory for which the meaning of events, entities, and situations can be represented through Frames, schematic representations of scenarios that involve participants, objects, and conceptual roles, known as Frame Elements. A Frame can be seen as a shared mental scenario or experience that people automatically activate to understand a word or expression. It provides the background knowledge needed to interpret who is involved, what happens, and how the situation normally unfolds. Each frame is evoked by a set of Lexical Units (LU), which may be a single word or a multiword expression, defined as the pairing of a lemma with a specific meaning.

For instance, interpreting the verb buy in the sentence 'Daniel bought a new car' requires a cognitive model/mental representation of commercial exchange. In this sentence, bought is the LU that evokes the Commerce_buy frame, whose corresponding FEs are Buyer (Daniel) and Goods (a new car).

By contrast, in a sentence such as 'Daniel really buys anything you tell him', the verb buy does not refer to the concrete act of purchasing an object, but to the metaphorical meaning of believing. Here, despite its identical linguistic form, 'buy' can be distinguished from the previous example as it evokes a different frame, Fall_for, which involves a distinct set of FEs compared to the commercial scenario.

This example is to show that each meaning of a word can activate a different frame, and that polysemous words (words with same form and different meanings) can be distinguished according to the frame they trigger. In essence, Frame Semantics views word meanings as understood in relation with the broader situations or experiences to which these words belong, and each frame includes characteristic roles, such as who acts, what is acted upon, and with what outcome.

Food beyond consumption

We are interested in capturing food from an historical perspective as well as considering any conceptualization of food which goes beyond its consumption as a source of nutrition. The purpose is to represent food in its realizations with a focus on its uses, manipulation and purposes different from nutrition. We express this through an adaptation of Frame Semantics and relations among frames as illustrated in Fig. 2. In the following paragraph, we present the characterisation of each frame through the specific semantic relations and their annotation process. These relations are used only to illustrate the different perspectives and uses through which food is conceptualised. The annotation, however, focuses exclusively on frames, and not on the relations between them. In Fig. 1 (from now on, scheme), we do not represent all the frame elements of each frame but just their Lexical Unit (the frame trigger word) and the Frame Element role which the food entity (FOOD_LU of the FOOD frame) acquired in this frame. For instance, in the sentence "boil potatoes gently", potatoes are both the Lexical Unit of the FOOD frame (namely, the food entity FOOD_LU) and the Produced_Food Frame Element in the COOKING_CREATION frame, according to boil which is the COOKING_CREATION Lexical Unit.

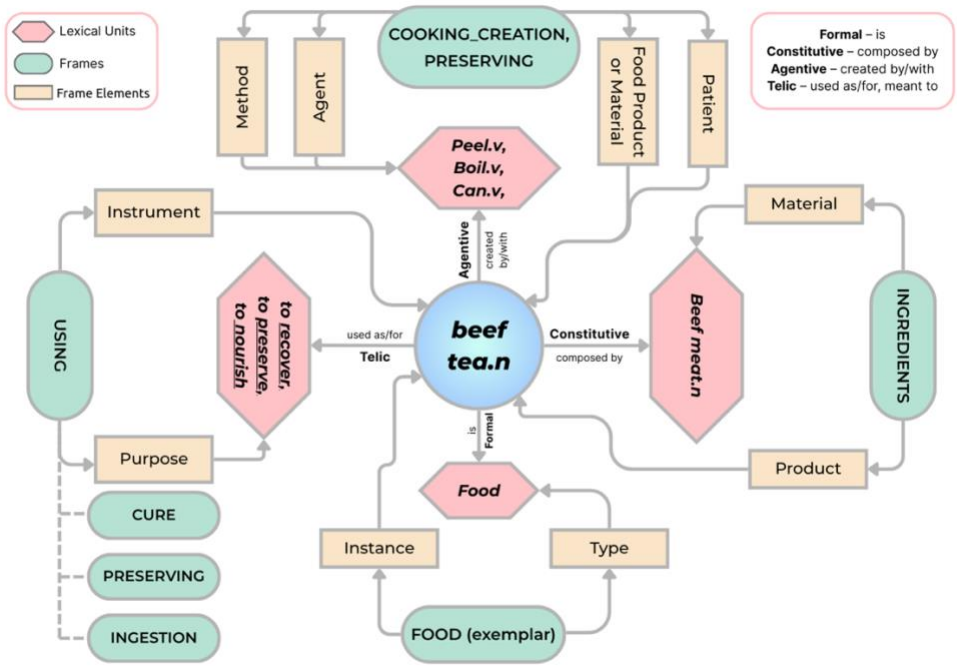


Figure 1: Schematic overview of the TRIFECTA Food Frames

Annotation

The annotation of foods and their uses with INCEpTION always begins with the identification of the Lexical Unit (LU) specific to the frame. The LU represents the key word or expression that triggers the frame: for example, for the Food frame, the LU is any term that refers to a food item (apple, soup, meat, etc.). The first step of the present annotation is the identification of the Lexical Unit of the frame Food. The standard definition of food in the Oxford English Dictionary is "any nutritious substance that people or animals eat or drink in order to maintain life and growth" and serves here as a baseline for identifying food. But, we do not restrict our notion of food to its act of consumption. For instance, substances that are normally ingested as nourishment but can also be applied externally, such as on the skin to treat a wound, are included in our conceptualisation of food.

In other words, we consider food as any substance typically used for nutrition, even when it is employed for non-ingestive purposes. The annotator has to select the word and choose the correct label Food_LU which stands for Lexical Unit of Food. The annotator has to annotate just the word referring to the food without the article, unless the article or any other element is indispensable to define the entity.

Important: each frame requires its own specific LU. Identifying the LU is always the fundamental prerequisite before proceeding with any frame annotation. For example, in a sentence like "I like apples", apples is a FOOD_LU, but it is not an INGESTION_Food_Ingestibles if there is no corresponding INGESTION_LU. In contrast, in "I like to eat apples", apples functions as both a FOOD_LU and an INGESTION_Food_Ingestibles, because eat serves as the INGESTION_LU.

Once you identified the LU, the possible frames which are "activated" are:

FOOD: entity food → IS

The Food frame is activated by a LU of Food, so any entity which is an exemplar of food (potato IS an exemplar of food). This Frame is defined in FrameNet as words referring to items of food, and it is considered the exemplar frame in this schema. This means we see the LUs of the frame as the starting point to activate all the possible relations with the other frames. For instance the word 'beef tea' in the flowchart is a LU of the Food frame. The Food frame consists of Food (LU), and the frame elements Constituent_parts, Descriptor, and Type (which is included in Descriptor in our guidelines). In addition there are negative, positive, and neutral annotations for Descriptors and 2 FE from Measures frame. The final frame in our schema is then structured as:

Food [FOOD_LU]: corresponds to the LUs [annotate the single food entity without the determiner];

1. Constituent_parts [FOOD_Constituent_Part]: a part of the Food_LU [Always annotate the full relevant span rather than just a single word. Example: In the sentence "a banana with a thick peel", annotate the entire span with a thick peel rather than only peel];
2. Descriptor [FOOD_Descriptor]: a characteristic or description, as well as a particular Type of the Food_LU; Descriptor includes: intensity, volume/reach, duration, state, character, hedonic characteristics. They can be (1) Positive [DESCR_Positive]; (2) Negative [DESCR_Negative]; (3) Neutral [DESCR_Neutral] [Usually, annotate single words as Frame Elements, including intensifying modifiers (e.g., very, really, much, most) in the same annotated span. E.g., in the sentence "this fish is really delicious", the Descriptor should be annotated as really delicious, not just delicious];
3. Count [FOOD_Count]: the number of units of the measured Food_LU [as a general rule annotate numerical expressions and avoid the annotation of a spoonful, a pinch, etc. with a as Count. In these cases, we consider a as part of the Unit Frame Element's annotation. Exception: a few should be annotated as Count. Potentially other quantifiers that imply a small, non-specific number, e.g., several, many, dozens, can also be treated as Count, depending on context];
4. Unit [FOOD_Unit]: the standardized quantity of Food_LU in terms of which other quantities of the same kind are measured [annotate Count and Unit as separate frames. Expressions like a spoonful, a drop, a pinch are annotated entirely as Unit, including the determiner a, rather than splitting into Count + Unit.].

Split of the FOOD frame. In the revised guidelines, the FOOD frame is split into three separate frames, so that food entity recognition can be evaluated as a separate task. The span rules above remain unchanged.

- **FOOD_ENTITY** marks every food entity [FOOD_LU].
- **FOOD_ATTRIBUTES** represents the quantity, measure unit and constituent parts of the food entity: Count [FOOD_Count], Unit [FOOD_Unit] and Constituent_Part [FOOD_Constituent_Part].
- **DESCRIPTOR** marks words that characterise food as Neutral [DESCR_Neutral], Positive [DESCR_Positive] or Negative [DESCR_Negative].

Examples: a basket of [oranges]FOOD_LU · [two]FOOD_Count [ounces]FOOD_Unit of orange [peel]FOOD_Constituent_Part · [excellent]DESCR_Positive [white]DESCR_Neutral wine; [rotten]DESCR_Negative eggs

INGREDIENTS: entity food → COMPOSED BY/OF

The Ingredients Frame is activated by any LU expressing any food material which is used to create a food product (potato mash is composed of potatoes). This Frame represents a Material used to make a Product, in this case any food which is used to create another. In our annotation guidelines, we kept the LUs and Frame Elements Material, and Product.

Core:

1. Material [INGR_Material_LU]: The entity which is used to make a Product [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU];
2. Product [INGR_Food_Product]: The entity or substance that is created out of the Material, in this case the food [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].

Notice that the annotator has to double annotate the FOOD_LU both as the frame LU and the specific FE of the other activated frame (in this case Product_Food of Ingredients). In the list of labels in INCEpTION, the FE which corresponds to the exemplar frame LU is marked with _Food label, for instance the FE of Ingredients, Product is Product_Food in our annotation schema.

COOKING_CREATION: entity food → CREATED BY/WITH

Unlike the previous two, these frames model events rather than entities. In this case, we consider three frames in the manipulative (created with) and agentive (created by) characterisations, namely COOKING_CREATION, APPLY_HEAT, and PRESERVING (boiled potatoes can be created by a cook and by boiling potatoes). These frames describe food and meal preparation and manipulation, including any type of manipulation performed, explicitly or implicitly, by an agent on the food, either to create or to modify it. For the COOKING_CREATION frame, we consider only the two core FEs, namely Cook and Produced_Food. While APPLY_HEAT focuses primarily on the process of handling and transforming the ingredients by heating it, COOKING_CREATION emphasises the edible entity resulting from the process. We aggregate these two perspectives under COOKING_CREATION, which covers a broader range of food manipulation verbs. Here, food annotated as Food_Material represents the raw food to which the process is

applied (as in APPLY_HEAT) while Food_Product represents the final product resulting from the manipulation (as in COOKING_CREATION).

- Cook/Peel/Grill the chicken (raw ingredient to which the process is applied)
- I Baked/Made an apple pie (a produced food is created from (raw) ingredients)

COOKING_CREATION [where we aggregated also APPLY_HEAT]:

- Cook [COOKING_CREATION_Cook]: The person who prepares the Produced_Food [annotate the Frame Element considering also the determiners and specifications (e.g., the chef, this cook, my dad, etc.].
- Produced_Food [COOKING_CREATION_Food_Product]: The Food_Product is the result of a culinary effort [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].
- Material_Food [COOKING_CREATION_Food_Material]: It is the raw food to which a culinary process is applied [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].

The COOKING_CREATION_LUs (with some of them borrowed by APPLY_HEAT) are all those words which evoke the cooking/culinary manipulation process. A non-exhaustive list is provided below:

bake.v, baking.n, barbecue.v, blanch.v, boil.v, boiling.n, braise.v, broil.v, broiling.n, brown.v, char.v, coddle.v, concoct.v, cook up.v, cook.n, cook.v, cooking.n, deep fry.v, fix.v, fry.v, frying.n, grill.v, grilling.n, make.v, melt.v, melting.n, microwave.v, parboil.v, plank.v, poach.v, preparation.n, prepare.v, put together.v, roast.v, roasting.n, roast.v, roasting.n, saute.v, scald.v, scorch.v, sear.v, simmer.v, simmering.n, singe.v, steam.v, steaming.n, steep.v, stew.v, stewing.n, toast.v, toasting.n, whip up.v

PRESERVING: entity food → relation in the scheme CREATED WITH

For the PRESERVING frame, we focus on the role of the preserved Patient. In our annotation, the Patient must be realised as food to trigger this frame, thus becoming Food_Patient, which is preserved to prevent deterioration, potentially using a Medium that can itself be food (Pickled cucumbers are created with the preserving process of pickling). While food can be annotated as either Patient or Medium, our primary focus in this realisation is on the Food-to-Patient transfer, which captures the manipulation aspect. Nonetheless, as we are also interested in cases where food functions as a preserving tool, its purpose-related realisation (in technical terms, the telic role), presented in the next section, includes instances where it acts as a preserving medium.

In this frame an Agent preserves a Patient (food) in order to prevent it from decaying. A Medium may be used.

1. Agent [PR_Agent]: The Agent is the person performing the intentional act that leads to the preservation [annotate the Frame Element considering also the determiners and specifications (e.g., Person Proper Nouns, my mom, his brother, etc.].
2. Medium [PR_Medium]: Medium is the substance in which the Patient might be submerged or treated to be preserved (e.g. vinegar, brine, salt, sugar). It also represents any means or instrument which is used to preserve such as: **container**: the vessel in which the Patient is kept (e.g. stone jars, those tins, this can), **method**: the preservation technique, when the frame is evoked by another LU (e.g. preserve them by smoking), or **instrument**: the device or agent used to preserve (e.g. a dehydrator, the sun) [annotate the Frame Element considering also the determiners and specifications (e.g., the smoke, this can, those tins, etc.].
3. Patient [PR_Food_Patient]: The Patient is organic or non-organic matter that undergoes preservation [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].

Example: [pickle]PR_LU the [cucumbers]PR_Food_Patient in [vinegar]PR_Medium

can.v, cure.v, dry.v, pickle.v, preservation.n, preserve.v, salt.v, tinned.a, smoke.v, buccan.v, bucan.v, pot.v, kipper.v, collar.v, souse.v

USING: prototypical frame for purpose → USED AS/USED FOR/MEANT TO

This last frame differs from the others in that it provides a prototypical structure for a set of frames that share the same purpose-oriented configuration, each capturing a different use or purpose of food. This frame represents a prototypical model whose structure mirrors that of other frames, but it should not be annotated in the texts. It serves only as a conceptual reference for frames describing the different uses or purposes of food. Essentially, it defines how elements related to the use or purpose of food are organized, and the concrete frames that are actually annotated are derived from this model. These frames refer to the specific function or intended purpose of an object, namely what it is meant to or what is used as/for (ginger is used to soothe sore throats). In the case of food, this dimension of telicity is encoded in the core FEs of the frame USING, which is taken here as a reference model for examining how use-driven

meanings of food are structured. These elements, recurring across the selected frames, support the conceptualisation of the various uses of foods.

Importantly, this prototypical frame can also guide the annotation when the intended use of the food is clear but there is no Lexical Unit explicitly listed for that function. For example, in the sentence "ginger is used for sore throats" (vs "ginger is used to alleviate sore throats" where alleviates is the CURE_LU), the core meaning is that ginger serves as a medicinal remedy, but there is no specific LU corresponding to CURE. In such cases, the construction "use for" functions as the LU for CURE and should thus be annotated as CURE_LU, allowing the annotator to capture the intended purpose even when it is not directly lexicalized in the LU list.

We used the core FEs of USING as a reference point to identify the shared FEs in the selected frames. Specifically, the relevant FEs are: Agent, the individual or entity using the instrument to achieve a goal; Instrument, the entity used by the Agent to fulfill their purpose which in our scheme is the FOOD_LU; and Purpose, which identifies the specific goal or reason for using the Instrument (here food), which in our scheme is represented by the specific LUs of the selected telic frames (e.g., CURE_LU, PR_LU, etc.).

For Using, the core FEs of this Frame are:

1. Agent: The Agent uses the Instrument to achieve their Purpose.
2. Instrument: An entity directed by the Agent that serves to bring about the Purpose of the Agent.
3. Purpose: This FE identifies the purpose for which an Agent uses the Instrument.

This frame has a list of lexical units which we provide below:

application.n, apply.v, employ.v, employment.n, exercise.v, operate.v, operation.n, operational.a, use.n, use.v, utilisation.n, utilization.n, utilise.v, utilize.v

The frames expressing the purpose-oriented characterisation are:

E1. CURE: entity food → USED FOR/AS, MEANT TO

Here, the purpose of food is to treat specific afflictions. It concerns a Healer treating or curing an Affliction (such as an injury, disease, condition, or pain) affecting a Patient, specifically when a food item is mentioned as a particular Treatment or Medication. We focus solely on the core FEs, and we aggregate Medication and Treatment into the single FE Food_Treatment. This choice reflects the fact that food is conceptually closer to Treatment, while not excluding its potential overlap with Medication.

The FEs are:

1. Affliction [CURE_Affliction]: It is generally the disease, injury or condition which is suffered by a Patient [annotate the Frame Element considering also the determiners and specifications (e.g., the plague, the worst fever I have ever had, the most painful sore throat I have ever experienced, etc.].
2. Body_part [CURE_Body_Part]: The Body_part is the specific area of the Patient's body which is treated [annotate the Frame Element considering also the determiners and specifications (e.g., on the body, his arms, my whole belly etc.].
3. Healer [CURE_Healer]: The Healer represents anyone who treats or cures the Patient [annotate the Frame Element considering also the determiners and specifications (e.g., Persian doctors, my mother, people who studied natural remedies, etc.].
4. Treatment/Medication [CURE_Food_Treatment]: The ingested, applied, injected, etc. substance designed to cure the Patient (in this case food) [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].
5. Patient [CURE_Patient]: This is the sufferer of the injury, disease or pain. However, Patient may not be expressed as a separate Frame Element but frequently occurs as the possessor of the Affliction [annotate the Frame Element considering also the determiners and specifications (e.g., sufferers from gout, children, my dad etc.].

Nested annotations. Within the CURE frame, nested annotations are allowed and must be annotated explicitly. Always annotate both the outer and the inner span. Nesting is allowed only among CURE frame elements.

- **Patients described through their affliction.** The whole span is annotated as CURE_Patient, and the affliction inside it as CURE_Affliction: [people who suffer from [gout]CURE_Affliction]CURE_Patient
- **Afflictions containing the body part they affect.** The whole span is annotated as CURE_Affliction, and the body part inside it as CURE_Body_Part: [pain in the [eyes]CURE_Body_Part]CURE_Affliction

Full example: [Dr. Smith]CURE_Healer prepares a [syrup]CURE_Food_Treatment to [ease]CURE_LU [pain in the [eyes]CURE_Body_Part]CURE_Affliction

A list of LUs is provided below:

alleviate.v, alleviation.n, curable.a, curative.a, curative.n, cure.n, cure.v, ease.v, heal.v, healer.n, incurable.a, improve.v, nurse.v, palliate.v, palliation.n, palliative.a, palliative.n,

rehabilitate.v, rehabilitation.n, rehabilitative.a, remedy.n, remedy.v, resuscitate.v, soothe.v, therapeutic.a, therapist.n, therapy.n, treat.v, treatment.n, subdue.v

E2. INGESTION: entity food → MEANT TO

In this frame, the purpose of food is its most common one: consumption. The frame formalises this by representing an Ingestor (who can be either a person or an animal) who consumes a food or drink item, the Ingestibles (here, Food_Ingestibles), an action that involves placing the item in the mouth with the intent of delivering it to the digestive system. Only in this frame, not only people but also pronouns referring to a person/animal like 'he', 'she', 'one', etc. should be annotated as Ingestor.

The core FEs of this frame that we considered in this study are:

1. Ingestibles [INGESTION_Food_Ingestible]: The Ingestibles are the entities that are being consumed by the Ingestor [annotate the single food entity without the determiner. In practice, re-annotate what is already labeled as FOOD_LU].
2. Ingestor [INGESTION_Ingestor]: The Ingestor is the person or animal eating or drinking [annotate the Frame Element considering also the determiners and specifications (e.g., Italians, my friend Abigail, populations of Asia, etc.].

The LUs are all those verbs or noun which refer to the act of eating and drinking

breakfast.v, consume.v, devour.v, dine.v, down.v, drink.v, eat.v, feast.v, feed.v, gobble.v, gulp.n, gulp.v, guzzle.v, have.v, imbibe.v, ingest.v, ingestion.n, lap.v, lunch.v, munch.v, nibble.v, nosh.v, nurse.v, put away.v, put back.v, quaff.v, sip.n, sip.v, slurp.n, slurp.v, snack.v, sup.v, swig.n, swig.v, swill.v, tuck.v

Annotation Rule for Titles in Culinary Recipes

There are specific annotation rules that apply to particular text types. In the case of culinary recipes, we decided to annotate the entire recipe title as INGR_Food_Product when the text that follows describes its preparation process. This decision is based on the structural nature of the recipe as a textual genre. Recipes are inherently oriented toward describing the use and combination of ingredients that come together to produce a defined final product, namely, the dish itself. In such cases, the title directly refers to the resulting food item that the subsequent instructions explain how to prepare. For this reason, we have chosen to formalize and systematize this annotation criterion. This rule does not apply indiscriminately to all titles. For example, it does not extend to recipes for remedies, medicinal preparations, or other non-culinary

formulations, where the outcome is not a food (such as recipes for ointments). The rule therefore applies exclusively to culinary recipes in the strict sense.

Labels

List of frames and associated labels

food IS

Frame: Food

- FOOD_Food_LU: Words referring to food items (i.e., LUs in the Food frame).
- FOOD_Constituent_Part: A component or a part of the FOOD_LU.
- FOOD_Descriptor: A characteristic or classification of the FOOD_LU, including both descriptive and typological information. It is further annotated for intrinsic polarity: (1) Positive [DESCR_Positive]; (2) Negative [DESCR_Negative]; (3) Neutral [DESCR_Neutral].
- FOOD_Count: The number of units in which the FOOD_LU is quantified.
- FOOD_Unit: The standard unit of measurement used to quantify the FOOD_LU (e.g., grams, liters).

Revision: these labels are now grouped into three frames: FOOD_ENTITY (FOOD_LU); FOOD_ATTRIBUTES (FOOD_Count, FOOD_Unit, FOOD_Constituent_Part); DESCRIPTOR (DESCR_Neutral, DESCR_Positive, DESCR_Negative).

food is COMPOSED BY

Frame: Ingredients

- INGR_Material_LU: The entity which is used to make a Product.
- INGR_Food_Product: The entity or substance that is created out of the Material, in this case the food.

food is CREATED BY/ CREATED WITH

Frame: Cooking_Creation

- COOKING_CREATION_LU Every word which expresses the action of cooking.
- COOKING_CREATION_Cook The person who prepares the Produced_Food.
- COOKING_CREATION_Produced_Food It is the result of a Cook's efforts.

Frame: Preserving

- PR_LU Every word expressing the act of preserving something from decaying.

- PR_Agent The person performing the intentional act that leads to the preservation.
- PR_Medium The substance in which the Patient might be submerged or treated with to be preserved, the container, the method or the instrument used to preserve it.
- PR_Food_Patient The organic or non-organic matter that undergoes preservation (in this case, food).

food is USED AS/USED FOR/MEANT TO

Frame: Using_Cure

- CURE_LU: Every word expressing the act of treating or curing someone.
- CURE_Affliction: It represents the injury, disease, condition, or pain felt by the patient.
- CURE_Healer: It represents anyone who treats or cures the Patient.
- CURE_Food_Treatment The ingested, applied, injected, etc. substance designed to cure the Patient.
- CURE_Patient It is the sufferer of the injury, condition, disease or pain.

Frame: Using_Ingestion

- INGESTION_LU Every word expressing the act of eating or drinking.
- INGESTION_Food_Ingestible The Ingestibles are the entities that are being consumed by the Ingestor.
- INGESTION_Ingestor The Ingestor is the person eating or drinking

INCEpTION

For instructions on how to use Inception, please follow this presentation (from slide 22) or refer to the official guidelines. For this task, the file to start annotating from is cook_books. To annotate, double-click on a word or drag the mouse to select multiple words. For span boundaries, please refer to the instructions provided in the description of each frame. If you happen to annotate or select an incorrect word or an incorrect span, simply select the tag and click Delete in the top-right corner (see screenshot below).

Annotation Workflow

In a given sentence, the annotator has to:

1. Identify the words referring to food and annotate them as FOOD_LU.

2. Annotate Frame Elements (FEs) related to each FOOD_LU in the FOOD frame.
3. Check whether the food item participates in another event or usage that activates another frame (e.g., CURE, INGESTION, PRESERVING), namely if any other LUs or food realisations are present. If so:
    a. Identify the FOOD_LU
    b. Annotate the verb (e.g., alleviates) - or any other argument in case of INGREDIENTS - as the LU of that frame (e.g., 'alleviates' and 'soothes' as CURE_LU).
    c. Re-annotate the food item with its new role in the frame (e.g., CURE_Food_Treatment for ginger) and then the rest of frame elements pertaining to the frame (e.g., CURE_Afflicition).

Annotation layers. Each frame is annotated in a separate layer. Frames are not mutually exclusive, so the same token can carry labels from several frames. For example, in "pickle the cucumbers in vinegar", cucumbers is both a FOOD_LU and a PR_Food_Patient, and vinegar is both a FOOD_LU and a PR_Medium. Within CURE, spans can also be nested (see E1. CURE).

More on FrameNet

Qualia Relations in FrameNet Brasil

Frame Semantics already embody a methodology whose aim is to represent the meaning of words especially through their context. Nevertheless, the representation of food in FrameNet is limited to considering food merely as an entity that can be ingested. But there are characterizations of food that do not entail its ingestion, as well as purposes that extend beyond mere consumption as a source of nutrition. For instance, when the purpose of food is clearly stated as a medicinal tool more than a nutritional item, which sometimes can entail not ingesting the considered food (e.g., a piece of meat applied on a bruise). To deal with this limitation, we resorted to the characterization of the different dimensions of a lexical item as presented in FrameNet Brasil. In this framework, the authors describe each lexical item through 4 main dimensions, named qualia roles. Each qualia role represents the different dimensions in which the meaning of a lexical item may be characterized.

The qualia structure encodes semantic dimensions essential to meaning, through four roles:

- **Formal (entity IS):** describes the basic category for the item and provides the information that distinguishes an entity within a larger set inside its semantic domain (e.g., banana is a food); Example: Ginger Ale is a plant-based drink.
- **Constitutive (entity IS COMPOSED BY):** expresses a variety of relations concerning the internal constitution of an entity (e.g., banana is formed by banana peel, pulp, etc.); Example: Ginger ale is made of carbonated water, ginger, and sugar.
- **Agentive/Manipulative (entity IS CREATED BY/WITH):** concerns the origin of an entity, its creator or its coming into being, every possible agent or methodology that led to the creation of the food entity we are considering (e.g., banana cake created by a cook, created by baking bananas, flour etc.) Example: The chef prepared the ginger ale using sparkling water.
- **Telic (entity IS MEANT TO/USED AS/USED FOR):** concerns the typical function or purpose of an entity, i.e., what the entity is meant to or used for (e.g., banana used as medicine). Example: Ginger ale is consumed to soothe nausea.

Consulting Frames in FrameNet

In FrameNet, there is a collection of nearly 1,200 frames for the English language. The repository is structured in such a way that it is possible to consult both the list of lexical units for the frames contained, and the frames themselves with their descriptions, list of LUs, and frame elements. As we have borrowed and re-adapted frames from FrameNet, we provide a brief overview of the type of information available in the repository. For those wishing to examine the frame element descriptions in greater detail and trace their sources, the repository can be consulted directly. If you are searching for a specific Lexical Unit and wish to identify the frames that contain it, you can refer to the Lexical Unit index. By entering the desired word in the search bar, you will be presented with the frames in which it occurs.

Alternatively, if you wish to examine a specific frame along with its description and the frames it is associated with, you can simply search for it in the frame list.